

XRPL Validator — Daily Engineering Report • 2026-05-05

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 4 commits (xrpl-validator 4)

Engineering changes

- **feat(validator): refuse to sign until 3 consecutive independent state-hash matches** (85cb113 , xrpl-validator) VALAUDIT Phase 3 (va-03), simple form. Highest-leverage security fix in the audit. Pre-this-commit, the validator signed network_hash_bytes for every ledger regardless of whether independent state verification had succeeded; the only difference between "VERIFIED" and "network" was a log label. Per the audit, the gate was tautological — a no-op security control.
- **fix(security): enforce 0600 on validator seed; fatal on bad mode or corrupt content** (032c75a , xrpl-validator) VALAUDIT Phase 4 (va-04) — seed hygiene. Three behavior changes hardening how the validator's signing seed is loaded/created at startup: 1. Permission check is FATAL on mode & 0o077 != 0 (group/other-readable) instead of warn-only. Operators ignored the previous warning; the audit's finding was that a key-material file at the wrong mode is a compromise-class issue, not an annoyance.
- **feat(dashboard): surface VALAUDIT Phase 3 gate + new diagnostic counters** (43f4bd1 , xrpl-validator) The validator has been growing observability for two weeks (silent + mutation divergence detection, RPC-fallback type breakdown, signing gate). The dashboard at halcyon-names.io was still rendering only the metrics from before all of that. This catches it up.

Docs and operations

- **chore(watch_engine): widen warmup threshold for VALIDATIONS SKIPPED color** (fe0242b , xrpl-validator)

Appendix — commits

Time	Repo	Commit	Subject
21:31	xrpl-validator	85cb113	feat(validator): refuse to sign until 3 consecutive independent state-hash matches
21:33	xrpl-validator	fe0242b	chore(watch_engine): widen warmup threshold for VALIDATIONS SKIPPED color
21:43	xrpl-validator	032c75a	fix(security): enforce 0600 on validator seed; fatal on bad mode or corrupt content
21:59	xrpl-validator	43f4bd1	feat(dashboard): surface VALAUDIT Phase 3 gate + new diagnostic counters